Michael Daniels

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Profile of Qualifications

- **Strategic Data Analytics Expert**: Proven track record in leveraging advanced analytics to drive strategic decision-making and operational excellence in dynamic environments. Adept at forecasting trends and optimizing processes to propel organizational growth.
- **Predictive Analytics & Modeling**: Proficient in employing predictive analytics and statistical modeling, skilled in using C++, Java, Python, R, SAS, and SQL to develop robust models that inform data-driven strategies.
- **Comprehensive Tool Mastery:** Proficient in a wide range of data analysis and manipulation tools along with expertise in data visualization using Tableau and Power BI. Additionally, skilled in using machine learning frameworks, notably PyTorch, and adept at implementing solutions on cloud platforms like AWS, Azure, and Google Cloud.
- **Impact through Data**: Exceptional ability to translate complex data insights into actionable business strategies, driving efficiency, growth, and competitive advantage. Experienced in aligning data analytics projects with strategic business objectives to maximize ROI.
- **Innovative Problem Solver:** Expert in leveraging advanced quantitative methods such as linear programming, game theory, and probability theory to devise innovative and actionable solutions to complex business problems, ensuring both creativity and practical applicability.
- **Project Leadership:** Proven history in leading and executing multiple concurrent analytics projects, consistently delivering superior outcomes within tight deadlines and budgets. Skilled in orchestrating cross-functional teams, cultivating a synergistic work culture that accelerates project milestones and success.

Career Highlights

- Spearheaded development and implementation of a new reinsurance data migration system for an emerging client company.
- Led the business analytics team that spearheaded the operation, uncovering $25M worth of missing claims spanning 3 years through strategic remediation efforts.
- Developed an internal analytics training program, educating our team of banking and insurance compliance specialists in the foundational principles of data science.
- Assisted with a research endeavor at New York Life, identifying and evaluating new ASAP reinsurance opportunities, resulting in the retention of premiums, and providing strategic insights for future growth.

Professional Experience

CARNEGIE MELLON UNIVERSITY (SEI) 2025 – PRESENT

DATA SCIENTIST; 2025

- Led software cost estimation and measurement efforts for complex, software-intensive and cyber-physical systems, developing quantitative models to assess cost, performance, and economic feasibility for government programs including DOD, DOE, and NOAA.
- Executed large-scale data processing, cleaning, and analysis, using Neo4J, SQL, Python, C++, R, and MATLAB to develop, validate, and refine statistical and machine-learning based methodologies that improved estimation accuracy and enabled early risk detection.
- Collaborated with cross-functional teams and senior stakeholders to align estimates with budget constraints, apply structured measurement frameworks, and communicate actionable insights.

W.R. BERKLEY CORPORATION 2024 – 2025

SENIOR DATA SCIENTIST; 2024

- Collaborated with stakeholders to gather and define data needs, translating business requirements into technical specifications for analytics and reporting.
- Partnered with cross-functional delivery teams to ensure data solutions aligned with business objectives and project timelines.
- Stayed abreast of data analytics trends, providing technical guidance and mentorship to junior team members in best practices and emerging technologies.
- Performed complex data analysis to derive actionable insights, supporting data-driven decision-making.

Professional Experience Continued

GEORGIA TECH RESEARCH INSTITUTE – ATLANTA, GA 2024 – 2024

GRAD ASSISTANT DATA SCIENTIST – MACHINE LEARNING ENGINEER; 2024

- Guided strategy formulation and provided recommendations for AI/ML Enterprise Adoption and Data Management Strategies.
- Identified improvements to business strategy based on emerging AI/ML potentialities and capabilities.
- Developed training materials to communicate AI/ML algorithms and services integration with the AF DCGS Enterprise.
- Interpreted and analyzed data, applying advanced algorithms in statistics, machine learning, LLM, NLP, and AI to solve problems and extract insights.
- Conducted thorough model validation and A/B testing plans, ensuring successful deployment by ML Engineers.

DELOITTE – REMOTE 2021 – 2023

MANAGER; 2023 – 2023

- Effectively manage, motivate, and coach high-performance teams throughout daily projects.
- Consistently ensure project activities align with state and federal regulations.
- Maintain the integrity and security of data throughout migrations.
- Offer the necessary resources and training for new analyst hires to succeed in meeting company objectives.
- Led the development of an internal comprehensive analytics training program, equipping team members with advanced data analysis techniques and tools to enhance decision-making capabilities.

RISK & FINANCIAL ADVISORY SENIOR CONSULTANT; 2021 – 2023

- Created risk mitigation strategies to boost efficiency based on the client's business, operational, regulatory, and technology needs and concerns.
- Maintained an understanding of key technological trends and workforce changes to assist clients with creating engaging and innovative technological improvements, including cloud computing, big data and analytics, process automation, artificial intelligence/machine learning, and other digital enhancements.
- Utilized innovative machine learning modeling capabilities to create compelling dynamic solutions to clients' unique problems.
- Assisted team members with developing and implementing revenue-generating sales initiatives, including lead-generation materials and sales proposals.

NEW YORK LIFE – NEW YORK, NY 2020 – 2021

ACTUARIAL ANALYST

- Collaborated with team members in optimizing, streamlining, and expediting current projects, including SOX premium testing, GMAD rate comparison tables, triangle reporting, and claims study charting.
- Utilized Prophet modeling systems to update life valuation ratings.
- Analyzed financial operations to determine profitability projections, assets, liabilities, and risks.
- Produced in-depth reports highlighting financial valuations, potential risks, existing processes, and actuarial/statistical research data.

NEW YORK LIFE – ATLANTA, GA 2017 – 2020

SENIOR SPECIALIST UNDERWRITER (LIFE AND MEDICAL)

- Conducted data analysis of current mortality standards and reinsurance business practices to protect company interests and adjust for updated mortality and general risk guidelines.
- Conducted thorough analyses of client applications, financial statements, medical documents, and lab results to verify eligibility and assign coverage for life insurance policies, annuities, and long-term care plans.

THE UNIVERSITY OF GEORGIA MATH DEPARTMENT – ATHENS, GA 2016 – 2017

RESEARCH ASSISTANT

- Conducted extensive reviews and synthesized research findings to identify knowledge gaps and opportunities for future research in ecology and mathematics.

Education & Professional Development

Master of Science in Computational Data Analytics/Statistics, Georgia Institute of Technology (2024)
Bachelor of Science in Applied Mathematics, Minor in Spanish, University of Georgia (2017)
Actuary Exams, Exams P, FM, IFM, and STAM
Yellow and Green Belt Certifications, Lean Six Sigma

Community & Professional Membership Involvement

Society for Industrial and Applied Mathematics: 2024-Present
American Statistical Association: 2026-Present
Humane Society Volunteer: 2023-Present

Honors & Achievements

Spanish Composition Contest (Earned "Excellent" Ranking), University of Georgia; 2015.
Eagle Scout; 2013